

POWERFACTORY

PowerFactory 2024

Technical Reference

Train, Train Line Model

ElmTrain, TypTrain

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The train model can be used for train simulation (moving trains a long at a line). AC (PH-E and PH-N model) and DC trains are supported. Further is the AC train model only available for unbalanced load flow calculation and the DC train model for balanced and unbalanced load flow.

2 Connections

The train could be connect in different ways

- Terminal: This is the *PowerFactory* standard connection and it is only used when calculating a load flow with disabled train simulation option (see load flow command, *Advanced Option* page → *Advanced*). When the train simulation option is enabled the connection of the train is ignored and the train model is isolated.
- Location and Position: For the location either a terminal (busbar) or a line with a corresponding position in km can be selected. When calculating a load flow (with enabled train simulation option) the train will be automatically connected to the terminal or to the line at the entered position.

2.1 Connection to a line

When the train is connected to a line, the line model must be configured as model for train simulation (see line element, *Basic Data* page → *Advance*: Line for train simulation).

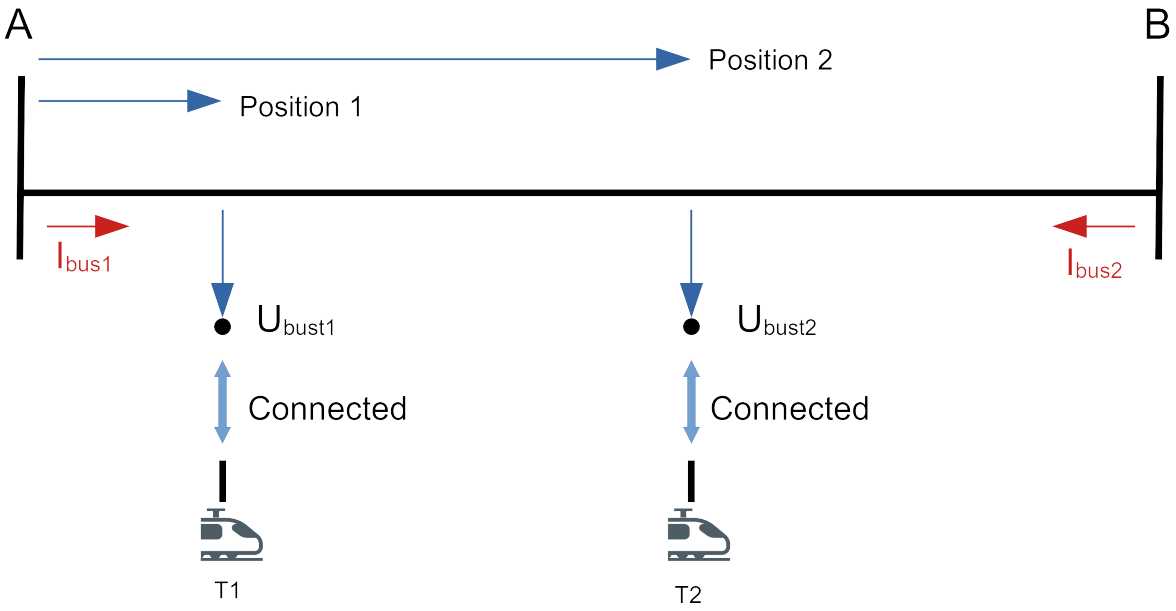


Figure 2.1: Example when two train models are connected at one line

The minimum resolution of train distance in m between two trains or between the terminal A,B and the train is shown in the line dialog *Basic Data* page → *Advance*.

When the distance between two trains are below the minimum resolution, both trains will be connected on the same internal node at position $pos = (pos(train1) + pos(train2))/2$.

In case of the position at the line is zero or below the minimum resolution, the train position will be internally increased to the minimum resolution.

The line model will be correspondingly adapted when one or more trains are connected. The maximum number of trains at a line is limited to $n = 10$ trains. The standard PI-model of a line will be then split into section as follow:

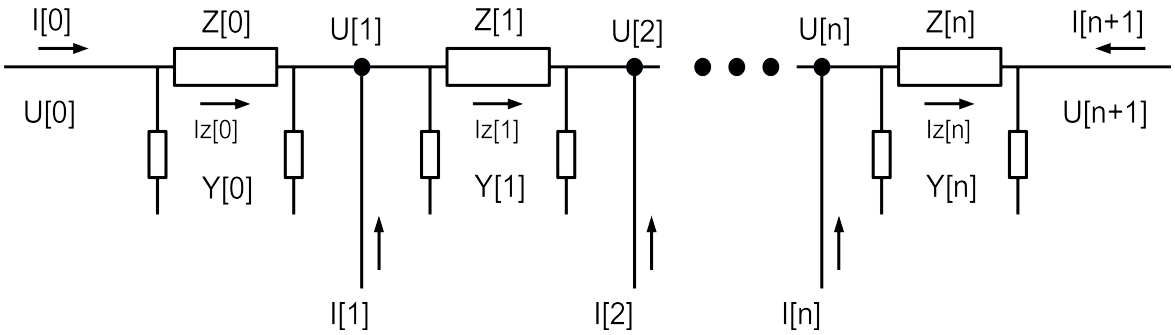


Figure 2.2: Train line model

where

$$
\begin{aligned}
Z[0] &= Z \cdot pos(1)/length \\
Y[0] &= Y \cdot pos(1)/length \\
Z[1] &= Z \cdot (pos(2) - pos(1))/length \\
Y[1] &= Y \cdot (pos(2) - pos(1))/length \\
&... \\
Z[n] &= Z \cdot (length - pos(n))/length \\
Y[n] &= Y \cdot (length - pos(n))/length
\end{aligned}
$$

and with:

PH-E model :

- Z is the complex impedance in Ω of the line ($R1 + j \cdot X1$])
- Y is the complex admittance in S of the line ($G1 + j \cdot B1$)
- $length$ is the length of the line in km
- $pos(1)$ ... $pos(n)$ are the train positions in km at the line

DC model :

- Z is the resistance in Ω of the line ($R1$)
- Y is the complex conductance S of the line ($G1$)

- $length$ is the length of the line in km
- $pos(1)$ ... $pos(n)$ are the train positions in km at the line

PH-N model :

In case of a PH-N model line, Z and Y are a complex impedance matrices:

$$[\underline{\boldsymbol{Z}}_{an}] = \begin{bmatrix} \underline{Z}_s & \underline{Z}_{pn} \\ \underline{Z}_{pn} & \underline{Z}_n \end{bmatrix} \qquad [\underline{\boldsymbol{Y}}_{an}] = \begin{bmatrix} \underline{Y}_s & \underline{Y}_{pn} \\ \underline{Y}_{pn} & \underline{Y}_s \end{bmatrix} \tag{1}$$

where the self impedance and admittance are:

$$\underline{Z_s} = R1 + j \cdot X1 \qquad \underline{Y_s} = G1 + j \cdot B1 \tag{2}$$

The values $\underline{Z}_n$, $\underline{Z}_{pn}$, $\underline{Y}_p$ and $\underline{Y}_{pn}$ of the neutral conductor can be directly used in (1).

For details of a single-phase line with neutral conductor, see Technical Reference of the Overhead Line Models.

Train model model :

The following quantities can be accessed from a train line model (*ElmLne*):

- $nconTrains$: no of trains at the line
- $conTrain : i$ connected train object i, with $0 <= i < n$ or via scripting function e.g. via DPL *line.GetVal(objectValue,'e:conTrain',i)*
- $conTrainPos : i$ position of the connected train i, with $0 <= i < n$ or via scripting function e.g. via DPL *line.GetVal(doubleValue,'e:conTrainPos',i)*

further after a successful load flow calculation the following calculation quantities are available:

- $c : loadingseg : i$ is the loading in % of the line segment i, with $0 <= i < n$
- $c : Isegi : i$ is the current in kA from side i (line begin) of the line segment i, with $0 <= i < n$
- $c : Isegj : i$ is the current in kA from side j (line end) of the line segment i, with $0 <= i < n$
- $c : loading$ is the max. loading in % from all line segment

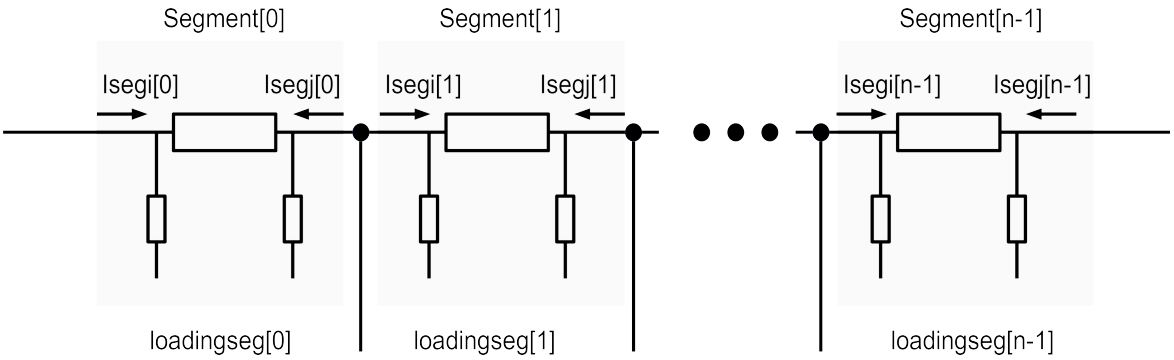


Figure 2.3: Calculation Quantities

2.2 Connection to a terminal

When for the location of the train a terminal (busbar) is selected, when it is like for all other *PowerFactory* elements e.g. load, the train model will be directly connected to the terminal.

2.3 No Connection

When for the location of the train nothing is selected, the train is still considered, but it is de-energized, so the train is on a kind of parking position.

3 Load Flow Analysis

The active power of the train model is split

1. Traction/Recuperation active power (*pini*) in MW
2. Auxiliary Active Power (*paux*) in MW

The active power at pantograph is:

$$Ppant = paux + pini \tag{3}$$

For AC models, the power factor can be entered for the traction/recuperation (alternatively as reactive power in Mvar) and for the auxiliary power. The reactive power at pantograph is:

$$Qpant = paux \cdot \sqrt{1/cos^2_{aux} - 1} + pini \cdot \sqrt{1/cos^2_{ini} - 1} \tag{4}$$

Note, then for the voltage dependency the option limitation acc. EN50641 is selected, only the power factor at pantograph can be entered.

3.1 Voltage dependency

3.1.1 Composite (ZIP)

For the *Composite (ZIP)* model, coefficient aP specifies the part of total active power which is constant power, coefficient bP specifies the part of total active power load which is constant current and coefficient $cP = 1 - aP - bP$ specifies the part of total active power which is constant impedance. The voltage dependency of the train is given in (5) and (6).

$$P = Ppant \left(aP + bP \cdot \left(\frac{v}{v_0} \right) + cP \cdot \left(\frac{v}{v_0} \right)^2 \right) \tag{5}$$

and only for AC models:

$$Q = Qpant \left(aQ + bQ \cdot \left(\frac{v}{v_0} \right) + cQ \cdot \left(\frac{v}{v_0} \right)^2 \right) \tag{6}$$

where v is the magnitude of the busbar voltage (in p.u.) and v_0 is 1 p.u. and $Ppant$, $Qpant$ is the entered power at pantograph, see equation (3) and (4). In case of a PH-N model is the voltage $v = u_A - u_N$, the phase-neutral voltage.

The *Composite (ZIP)* model is only considered in the load flow when the corresponding option *Consider Voltage Dependency of Loads* in the *Load Flow Calculation* command is enabled. If the option is disabled, the train model is considered as constant power model.

3.1.2 Exponent

For the *Exponent* model, coefficients eP and eQ specify the voltage dependency of the train according to (7) and (8):

$$P = Ppant \cdot \left(\frac{v}{v_0}\right)^{eP} \tag{7}$$

and only for AC models:

$$Q = Qpant \cdot \left(\frac{v}{v_0}\right)^{eQ} \tag{8}$$

where $Ppant$, $Qpant$ is the entered power at pantograph, see equation (3) and (4).

The *Exponent* model is only considered in the load flow when the corresponding option *Consider Voltage Dependency of Loads* in the *Load Flow Calculation* command is enabled. If the option is disabled, the train model is considered as constant power model.

3.1.3 Limitation acc. EN 50641

3.1.3.1 Traction

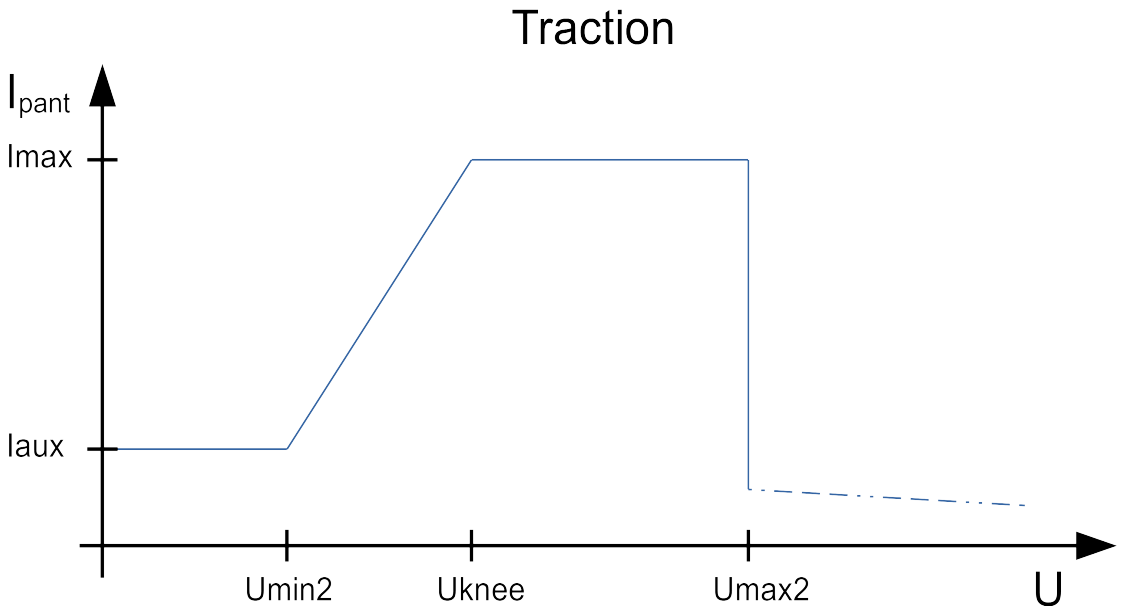


Figure 3.1: Current limiting in case of traction power

where for AC models:

$$Imax = \frac{paux + Ptrmax}{cosphi_{tr,max} \cdot Unom}$$
$$Iaux = \frac{paux}{cosphi_{aux} \cdot Umin2}$$

for DC models:

$$Imax = \frac{paux + Ptrmax}{Unom}$$
$$Iaux = \frac{paux}{Umin2}$$

and

- $Uknee = afactor \cdot Unom$
- $afactor$ is the knee point factor
- $Unom$ is the nominal voltage in kV of the train
- $Paux$ is the auxiliary power in MW
- $Ptrmax$ is the maximum traction power in MW

For AC models only :

The corresponding power factors ($cosphi_{tr,max}$ and $cosphi_{aux}$) are dependent on if the power factor is constant or power, voltage speed dependent. If the power factor is constant then: $cosphi_{tr,max} = cosphi_{aux} = cosini$ where the $cosini$ is the entered power factor at pantograph.
When a the power factor dependency is 'P, speed', the $cosphi_{tr,max}$ is dependent on the speed and the maximum traction power. The corresponding power factor from the entered table will be used.
The $cosphi_{aux}$ is also determined from the table, assuming that the traction power is zero.

The reactive power at pantograph is then:

$$Qpant = (paux + pini) \cdot \sqrt{1/cosphi^2 - 1}$$

In case of a dependency the power factor is either $cosphi = f(P_{traction,recuperation}, speed)$ or $cosphi = f(U_{pant}, speed)$. Where U_{pant} is the voltage at pantograph in kV and $P_{traction,recuperation}$ is the traction or recuperation power in MW.

3.1.3.2 Recuperation

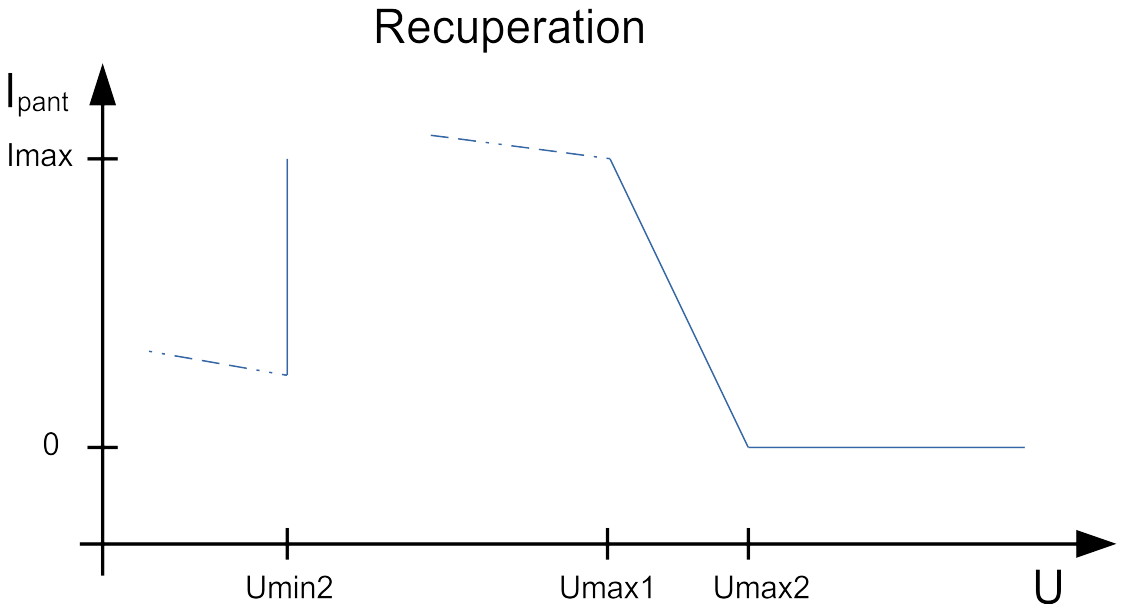


Figure 3.2: Current limiting in case of recuperation power

where for AC models:

$$Imax = \frac{paux + Premax}{cosphi_{re,max} \cdot Unom}$$

and for DC models:

$$Imax = \frac{paux + Premax}{Unom}$$

and

- $Umax1$ is Umax 1 in kV
- $Umax2$ is Umax 2 in kV
- $Umin2$ is Umin 2 in kV
- $Paux$ is the auxiliary power in MW
- $Premax$ is the maximum recuperation power in MW (must be smaller than zero)

For AC models only :

The corresponding power factor ($cosphi_{re,max}$ is dependent on if the power factor is constant or power, voltage speed dependent. If the power factor is constant then: $cosphi_{re,max} = cosphi_{aux} = cosini$ where the $cosini$ is the entered power factor at pantograph.
When a the power factor dependency is 'P, speed', the $cosphi_{re,max}$ is dependent on the speed and the maximum recuperation power. The corresponding power factor from the entered table will be used.

The reactive power at pantograph is then:

$$Qpant = (paux + pini) \cdot \sqrt{1/cosphi^2 - 1}$$

In case of a dependency the power factor is either $cosphi = f(P_{traction,recuperation}, speed)$ or $cosphi = f(U_{pant}, speed)$. Where U_{pant} is the voltage at pantograph in kV and $P_{traction,recuperation}$ is the traction or recuperation power in MW.

3.1.4 Constant current reduction

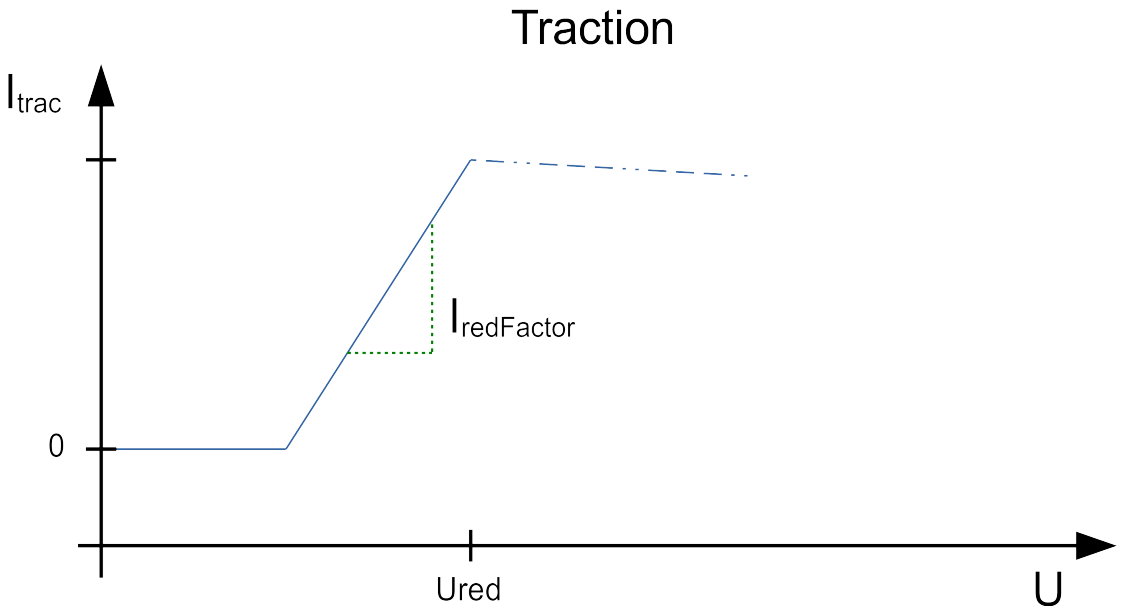


Figure 3.3: Constant current limiting in case of traction power

The traction power is reduced by a constant reduction factor (*IredFactor* in A/kV) when the pantograph voltage is below the voltage threshold $Ured$.

There is no limitation in case of recuperation.

For AC models only :

The reactive power at pantograph is then:

$$Qpant = paux \cdot \sqrt{1/cosphi_{aux}^2 - 1} + pini \cdot \sqrt{1/cosphi^2 - 1}$$

If no power dependency is enabled then $cosphi = cosini$. In case of a dependency the power factor is either $cosphi = f(P_{traction,recuperation}, speed)$ or $cosphi = f(U_{pant}, speed)$. Where U_{pant} is the voltage at pantograph in kV and $P_{traction,recuperation}$ is the traction or recuperation power in MW.

When the traction power is reduced by the limiting curve, the traction power factor is iteratively determined in case of the power factor is power, speed dependent.

3.1.5 Active power polygon

When the active power polygon is selected for the voltage dependency the active traction power and recuperation power (in %) dependent on the voltage (in kV) at pantograph can be entered in a table.

The percent values for the traction power are based on the max. traction power (*Ptrmax*) in MW. The values for the recuperation power are based correspondingly to the max. recuperation power (*Premax*) in MW. Note, the max. recuperation power must be entered as negative value.

3.1.5.1 Traction

Example:

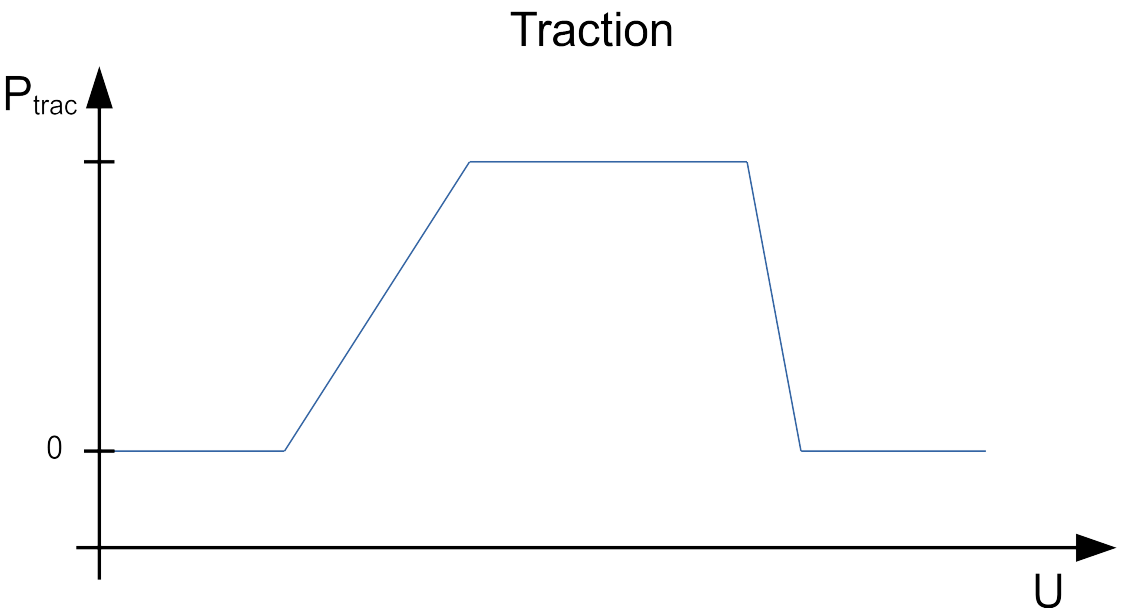


Figure 3.4: Active power polygon limiting in case of traction power

3.1.5.2 Recuperation

Example:

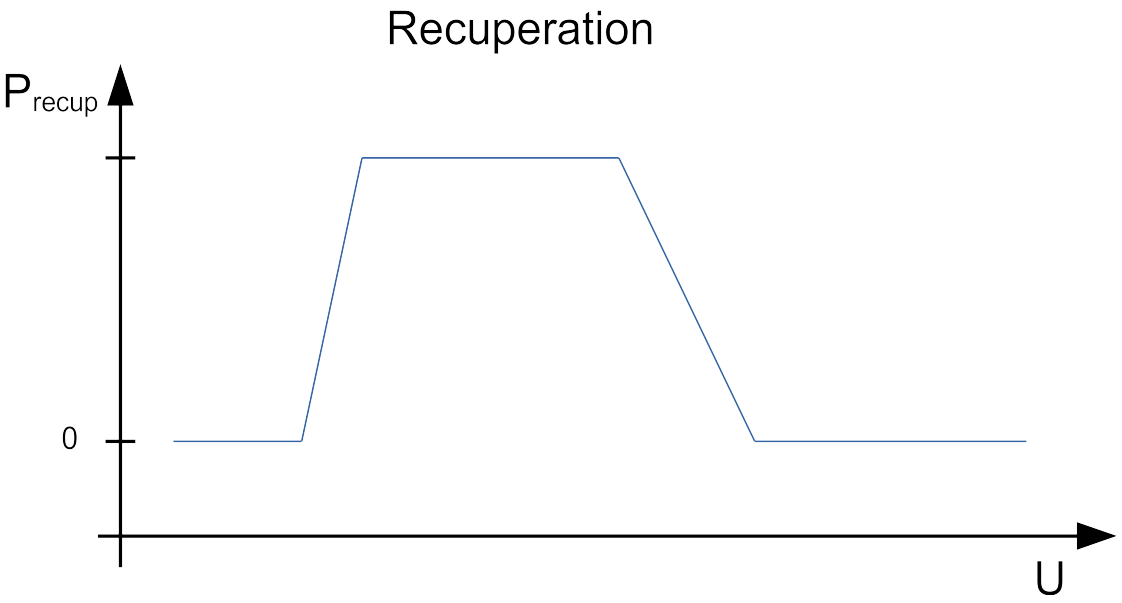


Figure 3.5: Active power polygon limiting in case of recuperation power

3.1.6 Current polygon

When the current polygon is selected for the voltage dependency the traction current and recuperation current (in %) dependent on the voltage (in kV) at pantograph can be entered in a table.

The percent values for both currents are based on the rated current in A of the train.

$$Irated = Sr/Unom * 1000$$

where

- Sr is the rated power in MVA
- $Unom$ is the nominal voltage in kV

3.2 Power factor dependency (only for AC models)

The power factor can be either defined as

- 'constant': constant power factor
- 'P, speed': power factor depends on the active power and the speed of the train
- 'U, speed': power factor depends on the voltage at pantograph and the speed of the train

Note, when the voltage dependency is according to EN 50641, the power factor dependency is the power factor of the pantograph. For all other, the power factor is defined only for the traction, recuperation.

3.2.1 constant

When no power factor dependency is defined the power factor is kept constant for the auxiliary power and for the traction/recuperation power.

3.2.2 P, speed

For the active power, speed dependency, the power factor can be entered in a table.

The active power points (in MW) and the speed (in km/h) must be entered. Note, a negative power factor is a Q injection, a positive power factor a Q consumption.

When for the train model the input mode 'P,speed' is selected, the power factor will be determine depended on the entered speed (linear interpolation between the entered 'speed' points). In the load flow model is then a $cosphi = f(P)$ characteristic used (smoothed linear interpolated).

3.2.3 U, speed

For the voltage, speed dependency, the power factor can be entered in a table.

The voltage points (in kV) and the speed (in km/h) must be entered. Note, a negative power factor is a Q injection, a positive power factor a Q consumption.

When for the train model the input mode 'P,speed' is selected, the power factor will be determine depended on the entered speed (linear interpolation between the entered 'speed' points). In the load flow model is then a $cosphi = f(U)$ characteristic used (smoothed linear interpolated).

3.3 Calculation Quantities

The following calculation quantities are available for the train model:

- $pini$ is the entered active traction/recuperation power in MW
- $cosini$[1] is the entered traction/recuperation power factor
- $paux$ is the entered auxiliary active power in MW
- $cosaux$[1] is the entered auxiliary power factor (not used when the voltage dependency is according to EN 50641)
- $Ptrac$ is the calculated active traction power in MW. Note $Ptrac$ differs from $pini$ in case of the traction power, current is reduce by the voltage dependency.
- $costrac$[1] is the calculated traction power factor. Note $costrac$ differs from $cosini$ in case of the power factor dependency is not constant. Also it is not used when the voltage dependency is according to EN 50641)
- pf_{coninj}[1] is the power factor consumption or injection flag for the traction power factor (1 = Q consumption, 0 = Q injection)
- $ilim$ is the current/active power limitation flag and is 1 when the traction power/current is limited, -1 when the recuperation power/current is limited and 0 when the power/current is not limited.
- $Pmism$ is the active power mismatch between the calculated power at pantograph and the entered active power (traction and auxiliary power) $Pmism = P - Ppant$.

[1] Only for AC models

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